9th Class • Punjab Board • PECTAA New Syllabus

Chapter 6: Data Science and Data Gathering

Complete solved exercise — data types, surveys, visualisation, Python, SQL, Power BI and original charts.

Chapter 6 • Complete official exercise • 7 MCQs + 9 short + 6 long questions

What this chapter is about

• **Data** are raw facts; information is data processed into a meaningful form. Qualitative data describe categories, while quantitative data measure amounts.
• A careful investigation defines the problem, collects suitable data, cleans and organises them, analyses patterns and communicates results through tables, charts and dashboards.
• Surveys, spreadsheets, databases, cloud tools, Python, SQL, Power BI and Tableau support different stages of gathering, storing, analysing and presenting data.
• Good data practice includes privacy, secure access, encryption, backups, cleaning, handling missing values and checking outliers rather than blindly trusting every result.

Source-check note

• MCQ 7's printed answer table gives (a) Data Cleaning. The verified process begins with **understanding/identifying the problem**, so the solved answer is (d). Both the printed key and the reason for correction are shown transparently.

Coverage: Chapter 6 corresponds to Unit 9 Data Science and Data Gathering in the twelve-unit textbook. The April 2026 nine-unit exercise is solved in book order.

A. Multiple Choice Questions

1. What is data?

(a) Processed information
(b) Raw facts gathered about things
(c) A collection of numbers only
(d) A list of observed events

Answer: (b) Raw facts gathered about things — Data are raw facts, observations or measurements; processing and interpretation turn them into useful information.

2. Which of the following is an example of qualitative data?

(a) Temperature readings in degrees Celsius
(b) Number of students in a class
(c) Favourite ice cream flavours
(d) Test scores out of 100

Answer: (c) Favourite ice cream flavours — Flavours are categories or labels rather than numerical measurements.

3. How can you organise data to make it easier to analyse?

(a) By writing it in long paragraphs
(b) By creating tables, charts, and graphs
(c) By storing it in random files
(d) By keeping it in a messy notebook

Answer: (b) By creating tables, charts, and graphs — Tables organise exact values, while charts and graphs reveal comparisons, patterns and trends.

4. What is the main purpose of data collection?

(a) To create random numbers
(b) To gather information to answer questions or make decisions
(c) To delete old data
(d) To format text documents

Answer: (b) To gather information to answer questions or make decisions — Data are gathered for a defined question or decision, not merely to accumulate numbers.

5. What is the primary purpose of data visualization?

(a) To generate random numbers
(b) To convert text into data
(c) To make data easier to understand by turning it into pictures
(d) To hide complex data

Answer: (c) To make data easier to understand by turning it into pictures — Visual forms such as bar charts and line graphs make patterns and comparisons easier to see.

6. Which tool is specifically designed for creating detailed and interactive visualizations?

(a) Microsoft Excel
(b) Google Sheets
(c) Tableau
(d) PowerPoint

Answer: (c) Tableau — Tableau is designed for detailed, interactive visual analytics and dashboards.

7. What is the first step in the data science process?

(a) Data Cleaning
(b) Data Analysis
(c) Data Collection
(d) Understanding the problem

Answer: (d) Understanding the problem — **Verified answer: (d).** The process first identifies and understands the problem, then collects and cleans data. **Source correction:** the printed answer table says (a) Data Cleaning, which conflicts with the chapter's process description.

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B. Short Questions

1. Write two differences between qualitative and quantitative data?

Qualitative data	Quantitative data
Describe qualities, categories or labels.	Describe numerical quantities or measurements.
Usually analysed by frequencies, proportions or themes.	Can be analysed with counts, means, ranges and other numerical methods.
Example: favourite ice-cream flavour or blood group.	Example: class strength or temperature in °C.

2. Which method would you use to collect opinions from a large group of people about a new school policy?

Use a **survey questionnaire**, preferably an online form for quick distribution and automatic tabulation. Ask a representative sample clear, neutral questions with suitable closed options and a small number of open comments, then summarise response frequencies.

3. What type of data is the number of students in your class?

It is **quantitative discrete data** because class strength is counted using separate whole-number values. A class can contain 30 students, but not 30.5 students.

4. Why is it important to organise data into tables or charts before analyzing it?

Organisation makes records easier to check, group and compare. Tables expose missing or inconsistent entries and preserve exact values; charts reveal patterns, trends and outliers that may be hidden in a long unstructured list.

5. Explain why data visualization is important.

Visualization turns values into visual forms so people can recognise comparisons, trends, relationships and unusual values quickly. It makes complex findings easier to understand, remember and communicate to decision-makers.

6. For which purpose do we use line graph?

A line graph shows how a numerical variable changes over a continuous interval, most often time. For example, it can display a student's average test score across six months or daily temperature over a week.

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7. How does Microsoft Power BI help in data visualization?

Power BI connects to data sources, helps clean and model data, and creates interactive charts, reports and dashboards. Users can filter and drill into results, refresh data and share dashboards for decision-making.

8. How can we use Python in Data Science?

Python can import, clean, transform, analyse and visualise data. Libraries such as pandas handle tables, NumPy supports numerical operations, Matplotlib and Seaborn create charts, and machine-learning libraries build predictive models.

9. What is SQL?

SQL means **Structured Query Language**. It is used to define and query relational databases—for example, SELECT retrieves records, INSERT adds them, UPDATE changes them and DELETE removes them, subject to permissions.

C. Long Questions

1. Explain the differences between qualitative and quantitative data.

The main distinction is whether observations describe **categories/qualities** or **numerical amounts**. The correct analysis and chart depend on this choice.

Feature	Qualitative data	Quantitative data
Meaning	Labels, names, qualities or categories.	Counts or measurements expressed as numbers.
Subtypes	Nominal: no natural order, e.g. city. Ordinal: meaningful order, e.g. poor/good/excellent.	Discrete: countable separate values. Continuous: measurable values on a scale.
Examples	Favourite sport, gender category, blood group, satisfaction level.	Number of books, marks, height, time, temperature.
Typical summary	Frequency, percentage, mode or themes.	Count, sum, mean, median, range and distribution.
Suitable visuals	Bar chart or pie chart for category frequencies.	Bar chart for counts; histogram, line graph or scatter plot where appropriate.
Caution	Numbers used as labels, such as ID codes, are still categorical identifiers.	A number is not automatically useful for arithmetic; check what it measures.

Classification examples

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Observation	Correct classification	Reason
Favourite ice-cream flavour	Qualitative — nominal	Flavours are names with no natural rank.
Small / medium / large shirt	Qualitative — ordinal	Categories have a meaningful size order.
Students present	Quantitative — discrete	Only whole counts are possible.
Height in centimetres	Quantitative — continuous	Height can take decimal measurements within a range.

2. Describe the process of conducting a survey to gather data about students' favourite extracurricular activities.

1. Define the objective: Decide exactly what must be learned—for example, the most preferred extracurricular activity among Class 9 students.

2. Identify population and sample: The population might be all 120 Class 9 students. Select a representative group, such as 30 students across all sections, rather than only friends.

3. Design the questionnaire: Ask one clear closed question with options such as Cricket, Football, Reading and Art. Add “Other” if needed and avoid leading wording.

4. Pilot and collect: Test the form on a few students, correct confusing wording, explain the purpose and collect responses anonymously where appropriate.

5. Validate and tabulate: Remove duplicate submissions, check that every response is allowed and count each category in a frequency table.

6. Analyse and visualise: Calculate frequencies or percentages and draw a bar chart. Compare categories without claiming more than the sample supports.

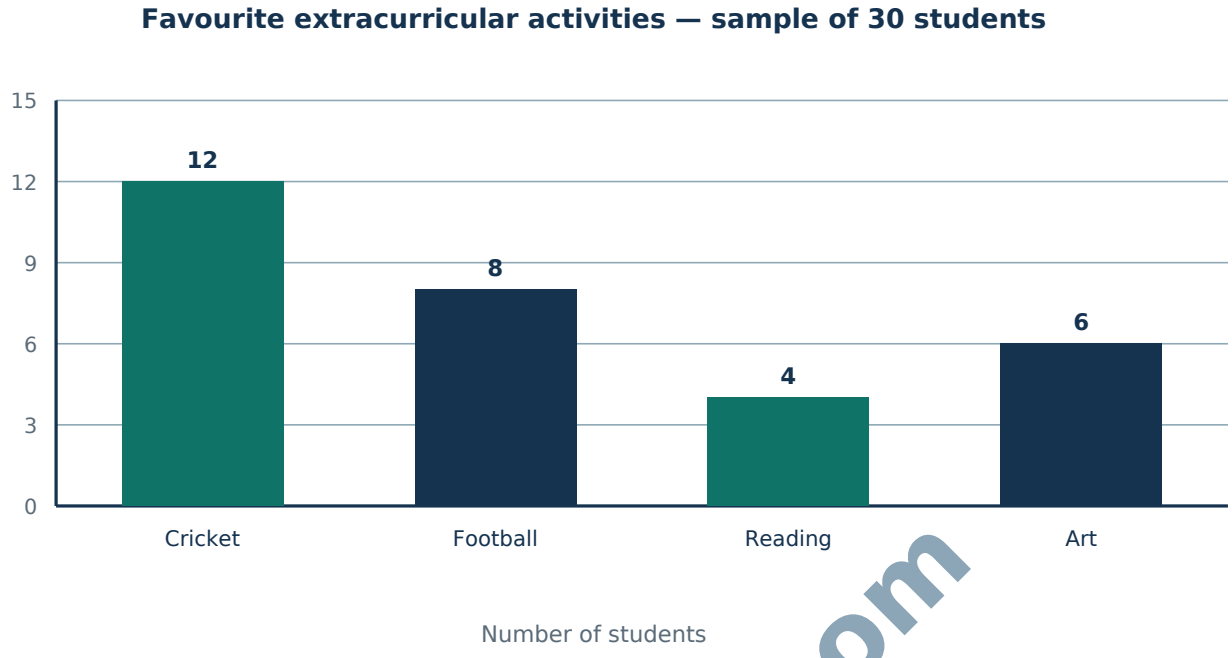
7. Report: State sample size, method, main finding and limitations; protect personal information and keep the raw data secure.

Model survey results (sample data)

Activity	Frequency	Percentage of 30
Cricket	12	40%
Football	8	26.67%
Reading	4	13.33%
Art	6	20%
Total	30	100%

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Figure 1 — Bar chart of the model survey



In this model, Cricket is the most frequent choice (12 of 30, or 40%). This conclusion describes the sample; a wider conclusion requires a representative sampling method.

3. Analyse the benefits and challenges of using digital tools like Google Forms for data collection.

Digital forms can make collection faster and more consistent, but their results still depend on access, question quality, sampling and careful checking.

Benefits	Challenges / controls
Forms are quick to create and share through a link, email or QR code.	People without devices or reliable Internet may be excluded; provide another response method when needed.
Responses are recorded automatically, reducing manual data-entry work and errors.	Duplicate, careless or false responses are still possible; use sign-in/limits only when justified and check the data.
Question types, required fields and validation improve consistency.	Poorly written or leading questions still produce biased results; pilot the form first.
Results can be exported to sheets for tables, statistics and charts.	Automatic totals do not interpret the result; choose suitable analysis and check categories and missing values.
Remote and large-scale collection saves printing, travel and time.	Privacy, consent, account security and data retention must be managed; collect only necessary information.

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Benefits	Challenges / controls
Updates can be made centrally and responses viewed soon after submission.	Platform dependence, access permissions and service outages can interrupt collection; keep an export/backup.

Digital tools improve the mechanics of collection but do not guarantee representativeness or accuracy. A good survey still needs a clear objective, suitable sample, neutral questions, ethical handling and careful checking before analysis.

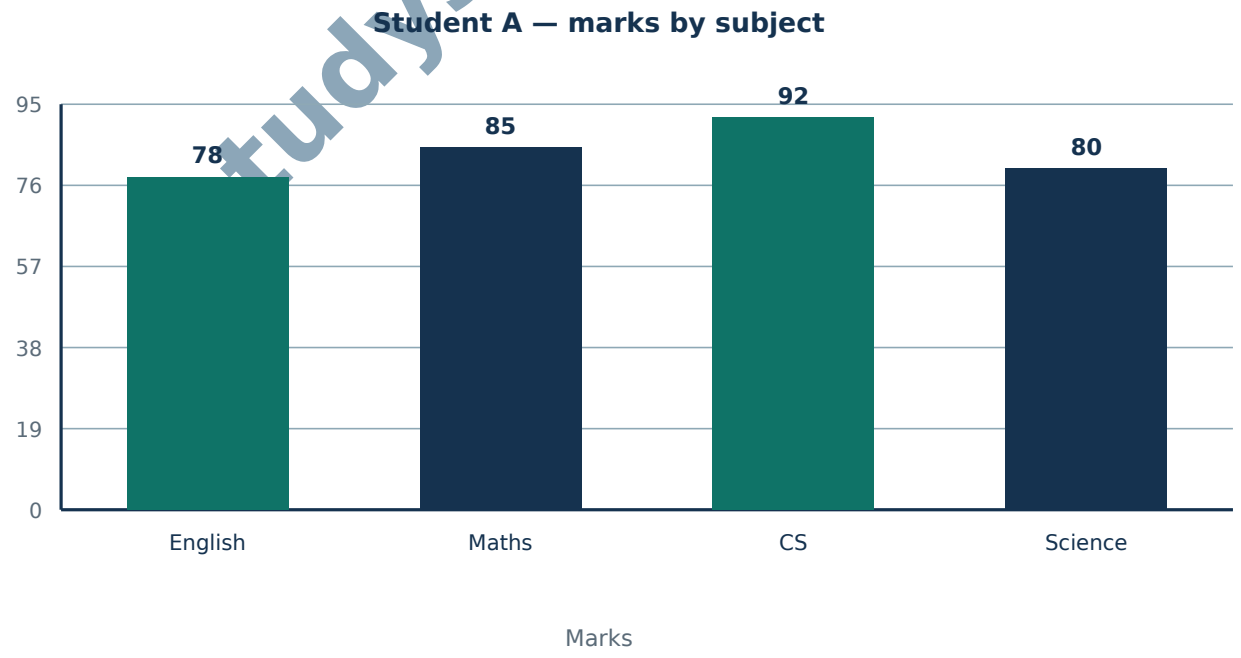
4. Explain the role of tables and charts in data analysis. Provide examples.

Tables organise raw or summarised values into rows and columns. They preserve exact figures, support sorting/filtering and make missing or inconsistent entries easier to detect. **Charts** encode values visually, making comparison, trend and distribution easier to interpret.

Example: marks table

Subject	Student A	Student B	Class mean
English	78	82	80
Mathematics	85	79	82
Computer Science	92	88	90
Science	80	86	83

Figure 2 — Student A's marks converted to a bar chart



• The table gives exact marks and allows Student A, Student B and the class mean to be compared directly.

• The bar chart immediately shows Student A's highest mark in Computer Science (92) and lowest in English (78).

• A **line graph** would be better for marks across consecutive terms because it emphasises change over time.

• A **scatter plot** would help investigate a relationship between two continuous variables, such as study time and marks, but would not by itself prove causation.

Tables and charts should agree with the source data and have clear titles, labels and units. A chart supports analysis; it does not replace checking the underlying values.

5. Discuss the importance and benefits of data visualization.

Data visualization represents values with visual elements such as position, length and colour. People can often see a difference or trend in a suitable chart faster than in a large numerical table.

• **Understanding:** It summarises complex data and makes the main message accessible to non-specialists.

• **Comparison:** Bar charts show which categories are larger or smaller and by approximately how much.

• **Trends:** Line graphs reveal growth, decline, seasonality and sudden changes over time.

• **Relationships:** Scatter plots reveal possible associations and clusters between numerical variables.

• **Anomalies:** Unusual bars or points draw attention to outliers and possible measurement or entry errors.

• **Communication and memory:** A clear visual gives a presentation or report a focused message that is easier to discuss and remember.

• **Exploration:** Interactive filters and dashboards let users test questions and inspect details without rewriting the dataset.

Good visualization must be honest and suitable: label axes, state units, use a fair scale, avoid unnecessary decoration and do not imply causation from correlation. Poor charts can mislead just as easily as poor calculations.

6. Why is it essential for businesses and decision makers to use charts, graphs, and dashboards?

Business decisions often depend on large, changing datasets. Charts and dashboards turn those data into current indicators that managers can compare against goals.

• **Fast situation awareness:** A dashboard can show sales, costs, stock, website visits and service response times together instead of through separate reports.

• **Trend and target monitoring:** Line charts show whether performance is improving, while target lines reveal whether a branch or product is behind plan.

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• **Comparison and resource allocation:** Category charts identify strong and weak products, regions or campaigns, helping managers direct stock, staff and budget.

• **Early warning:** Sudden changes and outliers can reveal falling demand, rising complaints, fraud or a failing process before the problem grows.

• **Interactive investigation:** Filters let a decision-maker move from a national summary to a city, product or time period and test possible explanations.

• **Shared communication:** A common visual report helps teams discuss the same evidence and record the reason for a decision.

For example, a retailer may see that online sales are rising while one branch's stock is falling. It can investigate and restock before losing sales. However, decisions still require **accurate, timely and representative data**, suitable definitions and human judgement; a dashboard is not proof by itself.

D. More Important Questions (from the text)

1. Differentiate between nominal, ordinal, discrete and continuous data and suggest a visualization for each.

Type	Meaning and example	Suitable visualization
Nominal	Unordered categories: favourite colour or city.	Bar chart of category frequencies; pie chart if parts of a whole are appropriate.
Ordinal	Ordered categories with unequal/unknown gaps: satisfaction low–medium–high.	Ordered bar chart; stacked bars for group comparisons.
Discrete	Countable separate values: students present each day.	Bar chart or dot plot of counts/frequencies.
Continuous	Measurable values with decimals possible: height or time.	Histogram for distribution; line graph over time; scatter plot for two variables.

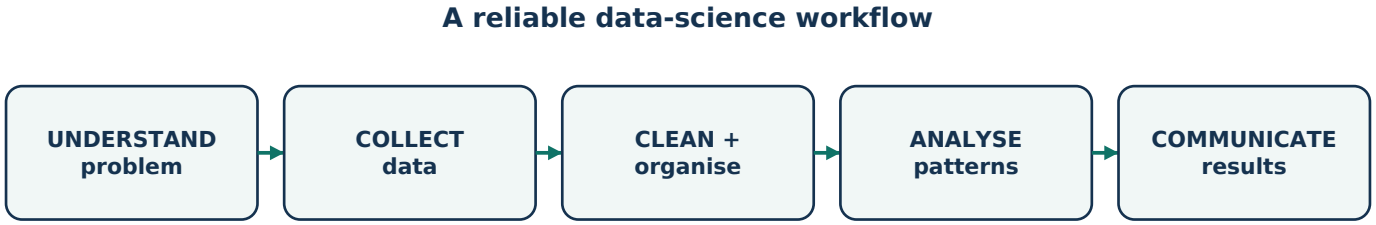
2. Why are data cleaning and outlier checking necessary?

Data cleaning corrects errors, standardises formats, handles missing values and removes unjustified duplicates. Without it, totals, charts and predictions may describe mistakes rather than reality.

An **outlier** is an unusual or extreme value that does not fit the general pattern. It may be a typing error, a different measurement unit or a genuine rare observation. Investigate the cause before changing or deleting it, and keep a record of every cleaning decision so the analysis remains reproducible.

3. Show the main data-science process as a flow diagram.

Figure 3 — Problem understanding comes before collection and cleaning



The process is often iterative: analysis may reveal missing data or a poorly framed question, requiring another collection or cleaning step. Results should include method, evidence, uncertainty and limitations.

4. How are the mean, median and mode calculated for the marks 78, 85, 85, 92 and 80?

Worked descriptive statistics

Ordered data: 78, 80, 85, 85, 92

Mean = (78 + 85 + 85 + 92 + 80) ÷ 5 = 420 ÷ 5 = 84

Median = middle (3rd) ordered value = 85

Mode = most frequent value = 85 (it occurs twice)

The mean uses every value, the median is the middle position after sorting and the mode is the most frequent observation. These measures can differ, especially when outliers are present; report the calculation and do not round before the final step.

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Points to Remember

Data: raw facts; **information:** processed, meaningful data.

Qualitative: nominal (unordered) or ordinal (ordered). **Quantitative:** discrete (countable) or continuous (measurable).

Workflow: understand problem → collect → clean/organise → analyse → visualise/communicate; iterate when needed.

Tables: exact values and checking. **Bar charts:** category comparison. **Line graphs:** change over time. **Scatter plots:** relationships.

Python: pandas/NumPy for handling/analysis and Matplotlib/Seaborn for charts. **SQL:** queries relational databases.

Power BI / Tableau: interactive reports and dashboards; automatic tools do not remove the need for critical thinking.

Outlier: unusual value to investigate, not automatically delete. Cleaning decisions should be documented.

Ethics: representative sampling, neutral questions, consent, minimum personal data, secure access and backups.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	c	3	b	4	b	5	c
6	c	7	d*						

* Source correction: Q7: printed table = (a). See the reason under the affected MCQ for the verified answer.

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How to attempt this chapter in the paper

• Classify the data before choosing a calculation or chart. ID numbers are identifiers, not automatically quantitative measurements.
• For a survey answer, include objective, population/sample, questionnaire, collection, cleaning, analysis, chart and conclusion.
• Label every chart with a title, categories, values and units; state when the data are a model sample rather than a real survey.
• For MCQ 7, remember the verified order: problem understanding comes before data cleaning; the printed key inconsistency is explicitly noted.